AQUAREV WATER · COMMERCIAL

Specifications & Installation Instructions

What the AquaRev water conditioning device is, where it goes on a commercial pool system, and how to install it. Written for engineers and contractors meeting the device for the first time. Companion to reference Drawing Set 100.

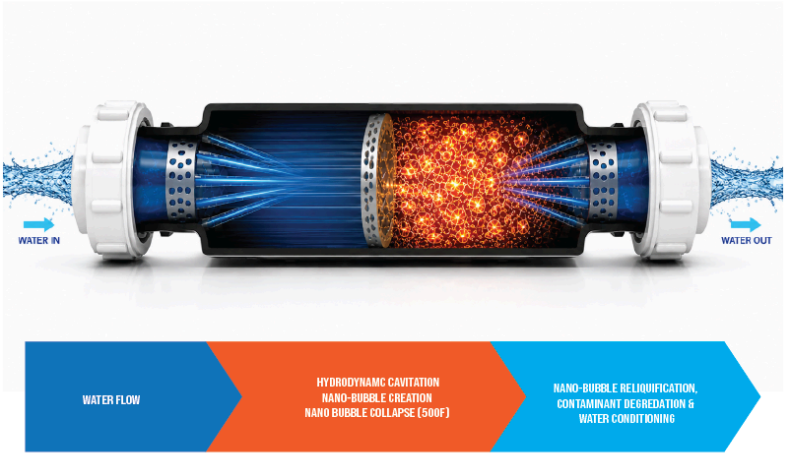
Revision A
September 2026

1 The device

A passive PVC vessel that mounts in-line on the pool return pipe. Water passing through its fixed internal shape undergoes hydrodynamic cavitation. Nothing to power, dose, clean, calibrate or replace. The process is described in Section 8, How it works.



The device as supplied: PVC body, a Schedule 40 union at each end.



Inside the chamber: water enters, hydrodynamic cavitation forms and collapses nano-bubbles, conditioned water leaves. Same graphic as the Data page.

LISTING Water Conditioning Device, certified to NSF/ANSI/CAN 50-2024 by IAPMO R&T, File No. W-12547. USPC® mark. Listed in compliance with the Uniform Swimming Pool, Spa and Hot Tub Code.

MATERIALS PVC body and fittings. NSF/ANSI 372 lead-free. California Prop 65 compliant. Made in the USA. No metal, so no grounding.

MOVING PARTS	None. No media, screens, lamps, cartridges, chemicals or electrical connection.
CONNECTION	In series on the existing return pipe using two Schedule 40 PVC unions supplied with the device. No bypass. No utility connection of any kind.
HEAD LOSS	2.6 psi rated, every model.
WARRANTY	Lifetime limited on the body. Requires professional installation and balanced water. Unions, O-rings and external piping excluded.

> **THE ONE RULE THAT NEVER CHANGES**
>
> The device supplements the pool's disinfection. It does not replace it. Free chlorine and pH stay inside the regulated range, tested and recorded exactly as before.

2 Where it goes and which model

One device per recirculating loop, always the last component before water returns to the pool.

Location

- On the return line, downstream of the pump, filter, heater, chemical injection points and sensor taps. Nothing is installed after it except existing pipe.
- As close to the pool return as the room allows. Any orientation: horizontal, vertical, diagonal.
- Existing equipment, valves, suction piping and drains are not touched. No dedicated pipe supports are needed; existing supports stay where they are.



Sample pump room. Flow runs left to right: from the pool, through pump, filter and chemical feed, then the AquaRev device last on the return. Plan view and dimensions on sheet M-101.

Model

Select by return pipe size, then confirm the loop's flow sits inside that model's range.

PIPE	MODEL	FLOW RANGE	L1	D1	STRAIGHT RUN
2 in.	KD2X4X2	44–127 gpm	20.25 in.	5.00 in.	28.25 in.
3 in.	KD3X6X3	112–273 gpm	24.00 in.	7.25 in.	32.00 in.
4 in.	KD4X8X4	240–480 gpm	28.19 in.	9.34 in.	36.19 in.
6 in.	KD6X12X6	440–1,100 gpm	40.00 in.	13.75 in.	48.00 in.

METRIC PIPE EQUIVALENTS

Outside North America, return lines are usually metric PVC to ISO 161 / EN 1452. Select the model by the nominal size, then confirm the pipe's actual outside diameter on site.

MODEL	IMPERIAL PIPE	SCH 40 OD	NOMINAL DN	METRIC PVC OD	FLOW M³/H	FLOW L/S
KD2X4X2	2 in.	60.3 mm	DN 50	63 mm	10–29	2.8–8.0
KD3X6X3	3 in.	88.9 mm	DN 80	90 mm	25–62	7.1–17.2
KD4X8X4	4 in.	114.3 mm	DN 100	110 mm	55–109	15.1–30.3
KD6X12X6	6 in.	168.3 mm	DN 150	160 mm	100–250	27.8–69.4

> **METRIC PIPE NEEDS TRANSITION FITTINGS**
>
> The supplied unions are imperial Schedule 40 PVC sockets and will not solvent-weld to metric pipe: 63 mm pipe is 2.7 mm larger than 2 in., 110 mm is 4.3 mm smaller than 4 in. Use a metric-to-imperial PVC transition coupling of the same nominal size on each side of the device, from the same cement system as the pipe. Confirm availability with AquaRev when ordering.

- **Clearance:** a straight run of L1 + 8 in. (4 in. clear each side to any fitting) and D1 + 4 in. around the body, with hand access to both union nuts. Full dimensions on sheet M-301.
- **Short run:** rebuild the run as an offset with four 90° elbows and put the device on the new leg (M-401, C).
- **Flow above the range:** two identical parallel branches with isolation valves, each device seeing half the flow (M-401, D). Confirm with AquaRev first.
- **Refer to AquaRev before ordering** when flow is within 10% of a range limit. Eight- and ten-inch units are available on request.

3 Before install day

An installation is a half-day plumbing task per loop. Nearly all the effort is in confirming the pipe, the flow, the space and the paperwork before anything is cut.

- [ ] **Site assessment record done**
  Return pipe outside diameter and material, flow from the meter or pump curve, clean-filter gauge reading, straight run available, isolation or drain-down method.
- [ ] **Pump margin confirmed**
  The device adds up to 2.6 psi.
- [ ] **AHJ determination in writing**
  Email reply or permit. A copy goes on site. See Implementation Guide §6–7.
- [ ] **Pool closure scheduled**
  Three to five hours per loop, including cement cure.
- [ ] **Device received and checked**
  Carton label photographed, serial recorded, two union halves and two O-rings present, IN and OUT identified.
- [ ] **Cut length measured from the unit**
  Dry-fit both union tails to the device and measure socket bottom to socket bottom. Write it down. Never cut to L1.
- [ ] **Primer and cement in date**
  ASTM F656 primer, ASTM D2564 cement, heavy-body for 4 in. and up.
- [ ] **Licensed installer, lock-out kit, ventilation**
  Locks and tags for the pump and every feeder on the loop. Fan or open door.

4 Installation

Eight steps, in order. Sheet M-302 draws each one; M-301 has the joint and cut-length details.

> **LOCK-OUT FIRST, EVERY TIME**
>
> The pump disconnect and every chemical feeder on the loop: chlorinator or salt cell, acid or CO_2 feed, and the controller's dosing outputs. Stopping flow with feeders live can concentrate chemical in the line.

01 Lock out and tag `M-302 · 01`

Pump off at its disconnect, locked and tagged. Every feeder and the controller's dosing outputs off, locked and tagged. Confirm nothing can restart.

02 Isolate and depressurize `02`

Close return isolation valves where fitted, otherwise open the filter drain. Open the filter air relief. The gauge must read zero. Bucket under the cut zone.

03 Measure, mark and cut `03 · 3/M-301`

Mark the recorded cut length on the return run after the last treatment component, 4 in. clear each side. Measure again. Cut square at both marks and remove the section.

04 Debur and clean `04`

Burrs off inside and out. Ends and sockets wiped clean and dry. A tail should dry-fit one third to two thirds of the socket by hand.

05 Prime and cement the union tails `05 · 2/M-301`

Nut onto the pipe first. Prime pipe end and socket, cement both, push fully home with a quarter turn, hold 30 seconds. Repeat the other end. Set the nuts so they turn by hand.

06 Set the device `06`

IN arrow toward the pump, label facing out. O-rings seated, faces clean. Confirm full insertion, then tighten each nut hand-tight plus a small snug. No wrench, no sealant, no cement on the threads.

07 Cure, then refill `07`

Wait the cement manufacturer's full cure for the size and temperature: typically 1 to 2 hours for 2 in., 2 to 4 hours for 3 and 4 in., longer when cold or damp. Close drains and refill the filter while waiting.

08 Restart, check and record 08

Pump lock-out off only. Start with the air relief open until clear. Inspect both unions and welds under pressure; a weep means reseat, not more torque. Gauge up to 2.6 psi more. Restore feeders last. Photograph the listing mark. Fill the record on M-501. Recheck at 24 hours.

5 After start-up

The device works from the first pump cycle. Keep dosing exactly as the health code requires and let the controller find any new demand on its own.

WHAT YOU SEE	WHAT TO DO
Filter loads faster, week one	Normal. Backwash or clean the cartridge as usual.
Controller doses less	Expected. Leave setpoints where the health code puts them.
Weep at a union	Depressurize, loosen, clean both faces, reseat the O-ring, retighten by hand. Never add torque, sealant or cement.
Gauge up more than 2.6 psi	Clean the filter first. If still high with a clean filter, check the device for debris.
Device suspected blocked	Lock out pump and feeders. Loosen unions, remove, turn 180°, run water backward 5 minutes, reinstall in the original direction. Then fix what let debris past the filter.
Winter	Winterize with the rest of the loop. The device drains through the unions.

No scheduled maintenance. Warranty claims go to water@aquarevwater.us with the install date and serial number.

6 Specification

For a purchase order, a contractor scope or a permit submittal. CSI MasterFormat Section 22 51 19, Swimming Pool Water Treatment Equipment. Renumber if the project manual uses another scheme.

PART 1 · GENERAL

1.1 **Work.** Furnish and install one AquaRev water conditioning device per recirculating loop, in series on the existing return line downstream of all treatment equipment, using the supplied Schedule 40 PVC unions. No modification to pumps, filters, heaters, feeders, controllers, suction piping, drains, shell or barriers. No utility connection.

1.2 **References.** NSF/ANSI/CAN 50-2024; IAPMO R&T Listing W-12547; NSF/ANSI 372; ASTM D1785, D2466, F656, D2564, D2855; local building and public-pool health codes; the AHJ's written determination.

1.3 **Submittals.** Site assessment record per loop; product data and listing; AHJ determination; installer licence; at closeout the completed installation record (M-501) with serial numbers and photographs.

1.4 **Quality.** Device bears the IAPMO R&T mark; no substitutions. Licensed pool contractor or plumber. Solvent joints per ASTM D2855 and the cement manufacturer, whose instructions govern where they differ from this document.

1.5 **Delivery and conditions.** Manufacturer's carton, stored indoors off the floor. Three to five hour closure per loop. Ventilation while cementing; no flame. Cure time per the manufacturer for size and temperature.

1.6 **Warranty.** Lifetime limited on the device body, conditional on professional installation and balanced water. Unions, O-rings and external piping excluded.

PART 2 · PRODUCTS

2.1 **Manufacturer.** AquaRev Water, aquarevwater.us. The listing carries the brand name HDC Products and the listee KM LLC dba KD Enterprises; cite it as issued.

2.2 **Device.** Passive in-line water conditioning device, PVC, lead-free, no moving parts or consumables, two Schedule 40 PVC union halves with O-rings supplied, label with listing mark, serial and flow direction. Models, ranges and dimensions per the table in Section 2.

2.3 **By the contractor.** ASTM F656 primer; ASTM D2564 cement, heavy body at 4 in. and up; Schedule 40 pipe and socket fittings of the return-line size for any reroute; lock-out devices and tags.

PART 3 · EXECUTION

3.1 **Examination.** Complete the site assessment record. Confirm the model's range brackets the measured flow and that pump margin covers 2.6 psi. Report non-PVC pipe, out-of-tolerance pipe or a run that cannot be made to fit before cutting.

3.2 **Installation.** Per Section 4 of this document and sheets M-301 and M-302: lock-out and zero pressure before opening the line; location per Section 2; cut to the field-measured length; joints per ASTM D2855; nuts hand-tight plus a small snug, no wrench; existing supports retained, none added; full cure before pressure.

3.3 **Field quality control.** Leak inspection of both unions and welds at operating pressure and again after 24 hours. Filter gauge increase of not more than 2.6 psi over the clean-filter reading. Feeders and controller restored last; setpoints unchanged.

3.4 **Closeout.** Assessment record, AHJ determination or permit and inspection, listing and warranty as issued, and the completed installation record in the project file.

7 Drawings & contacts



Reference Drawing Set 100 · Rev A

Eight ANSI B sheets drawn around a sample pool and spa pump room. Not site-specific; the same set serves every property.

OPEN THE DRAWING SET | DOWNLOAD PDF, 17 × 11

SHEET	USE IT FOR
G-001	General notes, symbols, abbreviations
C-101	Where the work sits on a property
M-101	Pump room plan: where the device goes on each loop
M-201	Flow diagram: what is upstream and downstream
M-301	Device dimensions, union joint, cut length, supports
M-302	The eight installation steps, drawn
M-401	Vertical runs, short runs, parallel devices, clearance
M-501	Materials, tools, closeout, installation record

Contacts and documents

SALES water@aquarevwater.us · (832) 979-6758

TECHNICAL AND WARRANTY aquarev@hdcproductsinc.com · (808) 628-0208

LISTING IAPMO R&T listing, File No. W-12547 · verify at pld.iapmo.org

RELATED DOCUMENTS Product Specification Guide 2026 and data page · Pump Room Implementation Guide Rev 1.1 (site assessment, approvals, permit determination)

8 How it works

Reference for Section 1. The process inside the chamber, from the Product Specification Guide.

As water passes through the AquaRev device chamber, the water undergoes hydrodynamic cavitation, temporarily converting from liquid into gas nano-bubbles. These cavitation bubbles implode inside the chamber and then re-liquefy before returning to the body of water. During the cavitation process, temperatures within the collapsing nano-bubbles can exceed 500°F destroying bacteria and removing heavy metals and other contaminants.

The system operates completely silently, produces no back-pressure, and contains no moving parts or electrical components. The result of this patented process is purified, cleaner, softer, healthier water and reduced costs.

Back to Section 1, The device · Lab reports and the Technical White Paper are on the Data page.